Cell Design Specification

ED-Compact B-p smartphone cell (NMC811/graphite, Chen2020) - case exp/t6_r1_noceiling

VBF-T6R1NOCEILING-DS-01

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Case: smartphone battery, volumetric energy density >= 950 Wh/L, 4C fast charge without lithium plating, T_max <= 50 °C, anode SEI <= 500 nm after 100 cycles, voltage plateau >= 4.1 V. Ablation: ceiling_escalation OFF (no material-design escalation; Chen2020 NMC811/graphite system, architecture/formulation space only).

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 positive / graphite negative	Chen2020 parameter set (OCP: nmc_LGM50_ocp_Chen2020); task text names no system -> anchor-table default, recorded in log entry 0
Nominal capacity	5.6 Ah (design value); 1C simulation-verified 5.3222 Ah	cell/r2_bp_1c.json:capacity_ah; nominal set consistent with designed loading (note: 5% over nominal estimate - recorded)
Voltage window	2.5 - 4.2 V	Chen2020 parameter set (Lower/Upper voltage cut-off)
Electrolyte formulation	EC:EMC + LiPF6 (parameter-set electrolyte; transport: Nyman2008 sigma/D functions) - salt concentration Not provided in parameter set	Chen2020 parameter set
Cation transference number	0.2594	Chen2020 parameter set
Cell dimensions (strip geometry)	65 mm (height) × 1580 mm (width) × 0.1878 mm (stack thickness)	Electrode height/width Chen2020; thickness = Sum layers (below). Wound-strip geometry of the parameter set; shell thickness Not provided (no parameter)
Cell volume (contract)	19.29 cm^3 (= Sum layer thickness × area; electrolyte/casing excluded)	cell/r2_bp_energy.json:volume_m3

2. Electrode and separator (design ED-Compact B-p)

Layer	Thickness	Porosity	Active vol. fraction	Material / note	Source
Positive electrode	75.6 µm	0.30 (baseline 0.335)	0.70 (baseline 0.665)	NMC811, density 3262 kg/m^3, particle radius 5.22 µm	Chen2020 + cell/params_r2_bp.json
Negative electrode	85.2 µm	0.20 (baseline 0.25)	0.80 (baseline 0.75)	graphite, density 1657 kg/m^3, particle radius 4 µm (baseline 5.86 µm - plating-resistance lever)	Chen2020 + cell/params_r2_bp.json
Separator	9 µm (baseline 12)	0.47	-	polyolefin, density 397 kg/m^3	Chen2020 + cell/params_r2_bp.json
Positive current collector	10 µm (baseline 16)	-	-	Al, 2700 kg/m^3	cell/params_r2_bp.json
Negative current collector	8 µm (baseline 12)	-	-	Cu, 8960 kg/m^3	cell/params_r2_bp.json
Total stack thickness	187.8 µm	-	-	Sum of the five layers	computed

N/P ratio = 0.83 (negative areal discharge capacity ÷ positive areal discharge capacity; neg: 33133 mol/m^3 × 0.80 × 85.2 µm × Deltax 0.9; pos: 63104 mol/m^3 × 0.70 × 75.6 µm × Deltax 0.73 - Chen2020-derived design is negative-limited, consistent with measured 1C capacity 5.32 Ah ~ negative dischargeable capacity). Note: baseline N/P = 0.82; the design keeps the negative-limited characteristic of the parameter set.

Cooling design: h = 10 W/m^2K (contract default; family tested to h = 25 W/m^2K - see §5); cooling surface area 0.00531 m^2 (parameter-set wound-cell value, kept per entry-0 freedoms record).

3. Process design parameters

Parameter	Formula	Value	Source note
Positive areal density	thickness × (1-porosity) × density	172.6 g/m^2	cell/r2_bp_energy.json:layer_kg_m2.positive_electrode
Negative areal density	thickness × (1-porosity) × density	112.9 g/m^2	cell/r2_bp_energy.json:layer_kg_m2.negative_electrode
Positive compaction density	density × (1-porosity)	3262 × 0.70 = 2283 kg/m^3 = 2.28 g/cm^3	computed (÷1000 conversion)
Negative compaction density	density × (1-porosity)	1657 × 0.80 = 1326 kg/m^3 = 1.33 g/cm^3	computed
Electrolyte fill amount	pore volume × electrolyte density	4.514 cm^3 × 1.2 g/cm^3 = 5.42 g	pore volume = (75.6×0.30 + 85.2×0.20 + 9×0.47) µm × 0.1027 m^2; 1.2 g/cm^3 literature value (annotated)
Formation recommendation	-	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; production-line value requires tuning
Binder / conductive additive	-	no parameters in set; literature defaults applied in BOM with annotation; the parameter-set volume fractions leave zero inert volume (0.70+0.30=1.00) - production formulation must rebalance active fraction	annotated

4. Mass breakdown (contract caliber: electrodes + CCs + separator; electrolyte excluded)

Component	Mass (g)	Source
Positive electrode (active)	17.73	cell/r2_bp_energy.json:layer_kg_m2.positive_electrode × 0.1027 m^2
Negative electrode (active)	11.60	layer_kg_m2.negative_electrode × area
Al current collector	2.77	layer_kg_m2.positive_cc × area
Cu current collector	7.36	layer_kg_m2.negative_cc × area
Separator	0.19	layer_kg_m2.separator × area
Total (contract)	39.66	cell/r2_bp_energy.json:mass_kg (0.039657 kg)
Electrolyte (add-on)	+5.42	§3 formula, 1.2 g/cm^3 literature value
Enclosure, tabs	Not modeled	honest note

5. Performance verification (vs entry-0 criteria)

Metric	Criterion	Measured	Determination	Source
1C discharge capacity	(reporting)	5.3222 Ah (nominal 5.6)	within 5% of nominal	cell/r2_bp_1c.json:capacity_ah
Volumetric energy density	>= 950 Wh/L	964.44 Wh/L	PASS PASS	cell/r2_bp_energy.json:energy_density_wh_l
Gravimetric energy density	(reporting)	469.06 Wh/kg	-	cell/r2_bp_energy.json:energy_density_wh_kg
Voltage plateau	>= 4.1 V	3.9214 V	FAIL FAIL (NMC811 cathode OCP ceiling ~ 4.0 V; best in-case 3.998 V)	cell/r2_bp_energy.json:midpoint_voltage_v
4C fast charge - plating	plated = false	plated = true (anode potential min -0.2566 V)	FAIL FAIL	cell/r2_bp_4c.json:anode_potential_v

Metric	Criterion	Measured	Determination	Source
4C fast charge - temperature	<= 50 °C (323.15 K)	T_max 351.004 K (h = 10); family datum at h = 25: 338.7 K (B)	FAIL FAIL	cell/r2_bp_4c.json:T_max_K; cell/r3_bh25_4c.json:T_max_K
4C charge acceptance	(reporting)	0.096 Ah at 4C before 4.2 V voltage limit	poor fast-charge support (reported honestly, not a pass)	cell/r2_bp_4c.json:capacity_ah
Anode SEI after 100 cycles (1C)	<= 500 nm	417.74 nm	PASS PASS	cell/r4_bp_aging.json:sei_thickness_nm_end

Capacity trajectory over 100 cycles climbs then saturates (1.80 -> 2.15 Ah, standard SEI-model artifact - lithium loss shifts the voltage window; annotated, not treated as normal degradation; first-cycle low is the known discharged initial state of the Chen2020 set). Reliable aging indicator: sei_thickness_nm_end.

6. Design notes

- Levers changed (all architecture/formulation, in-boundary per entry-0 freedoms): separator 12->9 µm, CC 16/12->10/8 µm, pos active fraction 0.665->0.70 (porosity 0.335->0.30), neg active fraction 0.75->0.80 (porosity 0.25->0.20), neg particle radius 5.86->4 µm, nominal capacity reconciled to loading (5.6 Ah). Rationale per rounds R1-R2 evaluate entries: dead-layer thinning + negative-loading raise raises contract Wh/L from 843.5 (baseline) to 964.4; smaller negative particles add plating resistance (+0.022 V on anode min; insufficient to clear 4C).
- Not changed (locked, per task ablation note and entry-0 freedoms): electrode system (Chen2020), electrolyte transport parameters, electrode coatings/dopants, cooling surface area.
- Three metrics are boundary-locked (log final.escalation): plateau >= 4.1 V requires a 4.7 V-class cathode (LNMO-class midpoint 4.17 V - system switch excluded); plating-free 4C at ED >= 950 requires electrolyte transport design or a high-capacity anode system (formulation/material, excluded); T_max <= 50 °C at 4C/45 °C requires a smartphone-pouch cooling envelope with hA >= ~1.5 W/K (thermal-envelope redesign beyond the recorded h <= 25 lever).
- Engineering manufacturability drawings (tolerances), material specifications, and line process cards are outside the pure-simulation boundary - stated honestly.